

Understanding Data, Variation, and Sampling

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💡 Chapter box

Central question. Among the 395 students recorded in two Portuguese secondary schools, how strong is the linear relationship between recorded absences and final mathematics grade, as summarized by Pearson's correlation, and what visible features, coding sensitivity, and scope limits must accompany that summary?

Focal concepts. Represented observations, distributions, graphical and numerical summaries, relationships, and repeated-sampling variation.

Main evidence. The documented two-school student-performance record, in which one row represents one student and both focal variables are present for all 395 rows (Cortez, 2008).

Defensible scope. An exploratory, observational, local description. The record does not estimate an effect of changing attendance and does not by itself support generalization to other students.

Reading the focal associational argument is the **primary use** of this chapter; the evidence for it is analysed descriptively. Quality evaluation follows reading as a separate and dependent **secondary use**. For researchers, the discussion also supports selected stages of the construction workflow as an **auxiliary use**.

Chapter 1 is the sole conceptual prerequisite. Chapter 1a is optional and is not needed to read or use this chapter.

1 Start with the Question, Not the Statistic

A short school report might say:

Recorded absences were almost uncorrelated with final mathematics grades, so attendance does not matter for achievement.

The reported number in the first clause could be correct while the sentence as a whole is not. A coefficient summarizes only a specified feature of recorded values. The second clause changes the language from a local description to a much broader conclusion about what matters. Before accepting the sentence, ask four questions:

1. What exactly is being asked?
2. What do the available records represent?
3. What pattern is visible, and how should it be summarized?
4. What answer remains after design and scope constraints are retained?

These questions keep the study question visible while we learn descriptive tools. They also prevent the statistic from deciding the question after the result is known.

1.1 The focal question and estimand

The focal question asks how two measured variables vary together, so its mode is **associational**. It is analysed descriptively within a fixed analytic record. Its **estimand** is the declared linear association: the Pearson correlation between recorded absences and final mathematics grade for the defined group of 395 students. An estimand is the precisely specified quantity that would answer the question if it could be known. Here all 395 rows contain numeric entries; taking them literally, the calculated correlation, $r = .034$, is the value of that finite-record estimand under the recorded coding. No sampling error is attached to this complete-record description. Numeric completeness does not establish coding validity, and the zero-grade pattern examined below remains unresolved. Because the records are not established as a probability sample from a wider population, this value is not automatically an estimate of a wider-population correlation.

A graph remains necessary. Pearson's r compresses a two-variable pattern to one number and targets linear form. It cannot show whether observations form a curve or clusters, whether most absences occupy a narrow range, or whether a few observations strongly influence the result. The scatterplot and smooth serve as checks on the scalar summary. Covariance explains the unstandardized cross-product behind r . These tools serve one declared estimand; they do not create extra question–answer pairs.

A wider-population correlation or the causal effect of preventing absence would be a different estimand. Each would require additional support. A reader should therefore identify the question, estimand, and reported answer before deciding what a result means.

1.2 A compact study record

The data were adapted from a documented student-performance resource based on two Portuguese secondary schools (Cortez, 2008). Table 1 gives the minimum record needed before calculation.

Table 1

Compact record for the focal absence–grade description.

Feature	Recorded information	Why it matters
Case and unit	One row per student; 395 rows	One point in the scatterplot represents one recorded student
Focal explanatory variable	Absences, recorded as a count from 0 to 75	A large count is unusual but not automatically an error
Focal response variable	Final mathematics grade, recorded from 0 to 20	The bounded scale constrains interpretation
Numeric-entry completeness	Both focal fields contain numeric values in all 395 rows	The literal recorded-value denominator is 395; coding validity remains separate

Feature	Recorded information	Why it matters
Source and design	Observational school records; no probability sample or attendance intervention is established	Population and causal interpretations require support not supplied here
Analytic status	Exploratory local description	The graph and summaries describe the accessible record

The original source documents variables and study context, but a publicly available file is not the same as a random sample from an unlimited population. Nor does a complete pair of columns guarantee that the measurements capture every substantive meaning a reader might attach to the words *attendance* and *achievement*.

1.3 Reading support before evaluating it

Chapter 1 taught a reader to locate a question–answer pair, identify one argument with linked statistical and substantive layers, and trace Evidence, Justification, Inference, and bounded Claim. This chapter applies that approach. Sections 1–5 accumulate relevant component content and representative documentary traces. Only after the argument is visible will we ask selected quality-evaluation questions. Reading records what the accessible study says and where its support can be found; it does not silently repair or grade the study.

2 What the Data Represent

A dataset is not merely a rectangular collection of numbers. Each row, column, code, and missing entry represents a decision about what was observed and how it was recorded. These decisions determine what later summaries can mean.

2.1 Cases, units, and occasions

A **case** is the entity or occasion represented by one row. The **observational unit** is the unit on which information is observed. Sometimes they are the same; sometimes a person supplies several rows across visits, trials, or waves. If each of 120 students is recorded before and after a course, 240 rows do not represent 240 independent students.

In the focal record, one row represents one student and the student is the observational unit. There is one occasion per row. The identifier distinguishes rows; it is not a score to average. Table 2 makes those roles concrete.

Table 2

Selected fields from six records in the student-performance data.

Student ID	School	Absences	Final grade	Pass
U001	GP	6	6	No
U002	GP	4	6	No
U003	GP	10	10	Yes
U004	GP	2	15	Yes
U005	GP	4	10	Yes
U006	GP	10	15	Yes

2.2 Variable form and substantive role

A **variable** records a feature that can differ across cases. A **value** is the recorded entry for one case. Variable form and variable role answer different questions:

- **Form** asks how values are represented: quantitative, nominal categorical, or ordered categorical.
- **Role** asks what the variable does in the present question: explanatory, response, grouping, or identifier.
- **Scale and units** ask what a difference or ratio can mean.

Absences and final grade are quantitative in recorded form. Absences plays the explanatory role and final grade the response role for the focal associational question. Those roles do not establish causal direction. Another question could reverse them or use either variable only to define groups.

Table 3

Recorded form and substantive role are different properties of a variable.

Recorded feature	Form	Role in the focal question	Interpretive caution
Student ID	Nominal label	Identifier: links and checks records	Numerical-looking labels are not quantities
School	Nominal categorical	Possible grouping information	Codes name categories; their order has no arithmetic meaning
Absences	Quantitative count	Explanatory variable	Counts are bounded below and strongly right-skewed
Final grade	Quantitative bounded score	Response variable	A score is not identical to every meaning of achievement

Recorded feature	Form	Role in the focal question	Interpretive caution
Pass	Binary categorical	Derived grouping variable	Its definition depends on the documented cut point

2.3 Coding, missingness, and the denominator

Coding translates observations into analyzable values. A code can preserve or distort substantive meaning. The numbers 1 through 4 may label ordered study time categories without establishing equal-hour intervals. A value of zero may mean a genuine measured zero, an absence, a score below a threshold, or a missing-value code. The documentation must decide which.

The focal file contains 38 final-grade zeros. The UCI repository identifies the final grade as a value from 0 to 20 and reports no missing values (Cortez, 2008). Yet every zero-grade row also records zero absences, and 25 have nonzero second-period grades from 4 to 10. The accessible documentation does not explain that joint pattern. We retain the values for the primary as-recorded description, flag their meaning as unresolved, and use an explicit deletion sensitivity rather than silently treating them as either invalid zeros or missing values.

Missingness changes the set of cases supporting a summary. If 1,240 clinic visits contain 36 missing waiting times and 81 missing return statuses, the complete-case denominator is not determined by subtracting 36 and 81 unless the overlap is known. If 10 visits are missing both, the denominator with both variables observed is

$$1240 - (36 + 81 - 10) = 1133.$$

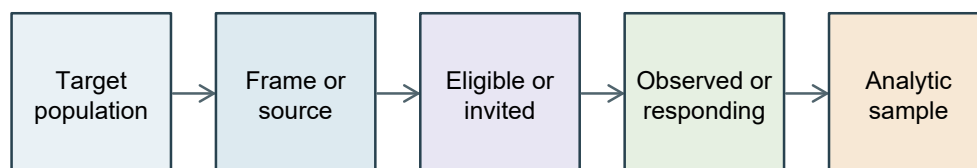
Always report the denominator that actually supports the graph, table, or coefficient. A source record, observed record, and analytic sample can differ. Both focal fields contain numeric entries in all 395 student rows. Even so, how the denominator works when values are absent is important because later studies will rarely be this tidy.

2.4 Follow the representation path

The representation path asks how the analytic cases came to stand for the people, occasions, or events named in a claim. Figure 1 shows the minimum sequence.

For the focal resource, the two schools and their recorded students provide a source and analytic record. The accessible documentation does not establish a probability sample from a wider target population. We should therefore leave unsupported links open instead of assuming that 395 cases represent all secondary students.

A small evidence audit before calculation is often enough:



Mark a step not applicable; do not invent a sampling step.

Figure 1: A representation path separates the intended population from the analytic sample.

1. What does one row represent?
2. Which variables, scales, roles, and units serve the question?
3. Which values need documentary checking?
4. Which cases enter each summary, and what is its denominator?
5. What path connects the analytic sample to the scope of the claim?

These are reading questions about represented evidence. They do not yet decide whether the study's support is adequate.

3 See and Summarize a Distribution

A **distribution** describes how a variable's values are arranged across cases. A useful beginner sequence is: identify the variable, display the observations, describe visible structure, choose center and spread, and say what the summary leaves unresolved.

3.1 See shape before reducing it

Figure 2 displays all 395 final grades. The graph makes several features visible: the score is bounded between 0 and 20, centered near 11, discrete rather than continuous, and has a visible group at zero.

Describe a distribution with four kinds of language:

- **Shape:** symmetry or skewness, clusters, gaps, and discreteness.
- **Center:** where a typical or balancing value lies.

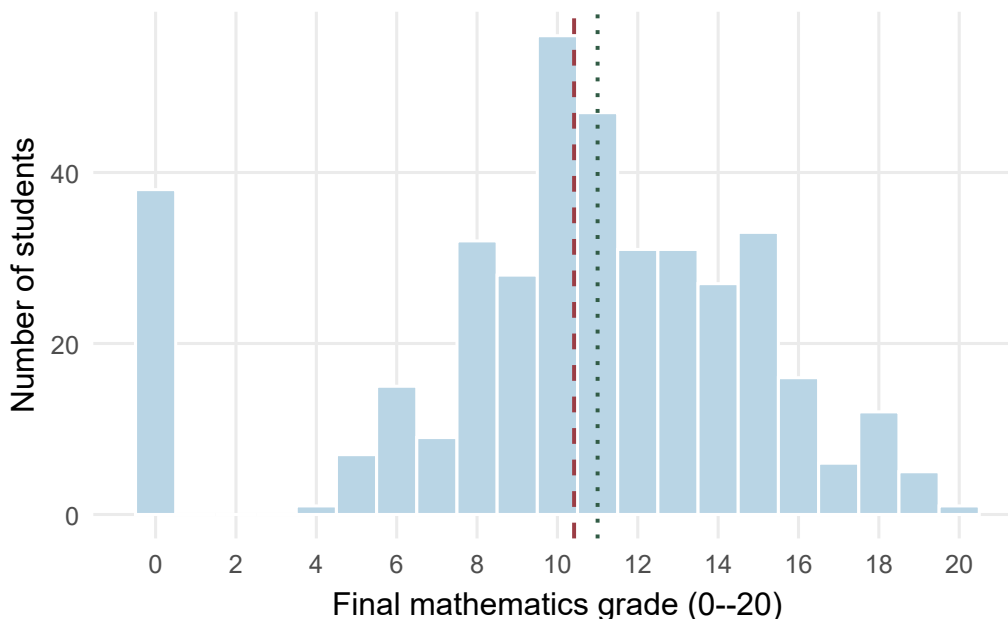


Figure 2: Distribution of final mathematics grades for the 395 recorded students.

- **Spread:** how widely observations vary.
- **Unusual observations:** values or patterns needing contextual checking.

An unusual value is not automatically a mistake. Its recorded meaning and provenance should be investigated before it is changed or removed.

3.2 Five summaries with distinct jobs

The main route uses exactly five numerical summaries: mean, median, sample variance, sample standard deviation, and interquartile range. They are a selected family, not the **five-number summary** of minimum, first quartile, median, third quartile, and maximum.

For observations $x_1, \dots, x_n$, n is their number and x_i is observation i . The symbol $\sum$ means “add the indicated terms”: $\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$. The mean, sample variance, sample standard deviation, and IQR are

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2,$$

$$s = \sqrt{s^2}, \quad \text{IQR} = Q_3 - Q_1.$$

The **median** is the middle ordered observation when n is odd and the average of the two middle observations when n is even. The first quartile Q_1 marks approximately the 25th percentile; the third quartile Q_3 marks approximately the 75th. They bracket the middle half of the distribution. For the small hand examples, take the median of each ordered half, excluding the overall middle observation when n is odd. Software may interpolate between values instead; the large-data summaries here use R's default interpolation convention. These rules can differ slightly in small datasets.

Table 4

Five selected summaries and the questions they answer.

Summary	Interpretive job	Units	Main caution
Mean $\bar{x}$	Arithmetic balance point; center used by much later algebra	Original units	Can move strongly with skewness or extreme values
Median	Ordered midpoint	Original units	Does not describe spread or tail shape
Sample variance s^2	Sum of squared deviations divided by $n - 1$	Squared units	Magnitude is hard to read directly in original units
Sample standard deviation s	Typical scale of deviation around the mean	Original units	Can be influenced by extreme values
IQR	Width of the middle half, $Q_3 - Q_1$	Original units	Does not show what happens in the tails

To make the formulas concrete, consider four illustrative completion times: 1, 3, 3, and 5 minutes. The sum is $1 + 3 + 3 + 5 = 12$, so $\bar{x} = 12/4 = 3$ minutes. The middle two values are both 3, giving median 3. The lower-half median is $Q_1 = (1 + 3)/2 = 2$ and the upper-half median is $Q_3 = (3 + 5)/2 = 4$, so $\text{IQR} = 4 - 2 = 2$ minutes.

Table 5

Working from observations to squared-deviation spread.

Observation x_i	Deviation $x_i - 3$	Squared deviation $(x_i - 3)^2$
1	-2	4
3	0	0
3	0	0
5	2	4
Sum	0	8

Thus $s^2 = 8/(4 - 1) = 8/3 = 2.67$ squared minutes, and $s = \sqrt{8/3} = 1.63$ minutes. Keep the

unrounded variance while taking its square root. Variance measures spread on a squared scale; SD restores the original units.

Why $n - 1$? Deviations from their own mean must sum to zero. Here, once the first three deviations $-2, 0, 0$ are known, the last must be 2. In general, only $n - 1$ deviations can vary freely: one **degree of freedom** is used by estimating the center. This zero-sum constraint explains the lost degree of freedom, but is not by itself a proof of unbiasedness. Section 5 supplies the sampling-model justification after defining expectation.

For a fixed observed distribution, dividing the same squared-deviation sum by n instead gives its mean squared deviation ($8/4 = 2$ here). Both conventions can describe spread when clearly labelled. We use the conventional sample variance s^2 throughout; the name and denominator do not establish a random sample or justify population claims. Nor is $n - 1$ the finite-population correction discussed in Appendix C.

Table 6 applies all five summaries to final grade.

Table 6

Five numerical summaries of final mathematics grade.

Summary	Value	Units
Mean	10.42	grade points
Median	11.00	grade points
Sample variance	20.99	squared grade points
Sample standard deviation	4.58	grade points
IQR	6.00	grade points

The mean is 10.42 and the median is 11. The sample variance is 20.99 squared grade points, while the standard deviation is 4.58 grade points. The IQR is 6 grade points. These numbers summarize the recorded distribution; they do not make the score scale continuous, validate the zeros, or establish a probability model.

Omitting the 38 zero-grade rows only as a coding sensitivity leaves 357 records: mean 11.52, median 11, variance 10.42, SD 3.23, and IQR 5. These are diagnostic quantities, not replacement summaries.

3.3 Equal means can describe different empirical worlds

Figure 3 shows three constructed distributions with the same mean of 10. Their shapes and spreads are plainly different.

A center cannot replace a distribution. Report mean with standard deviation when that pair serves the question and the visible shape. Report median with IQR when resistant summaries

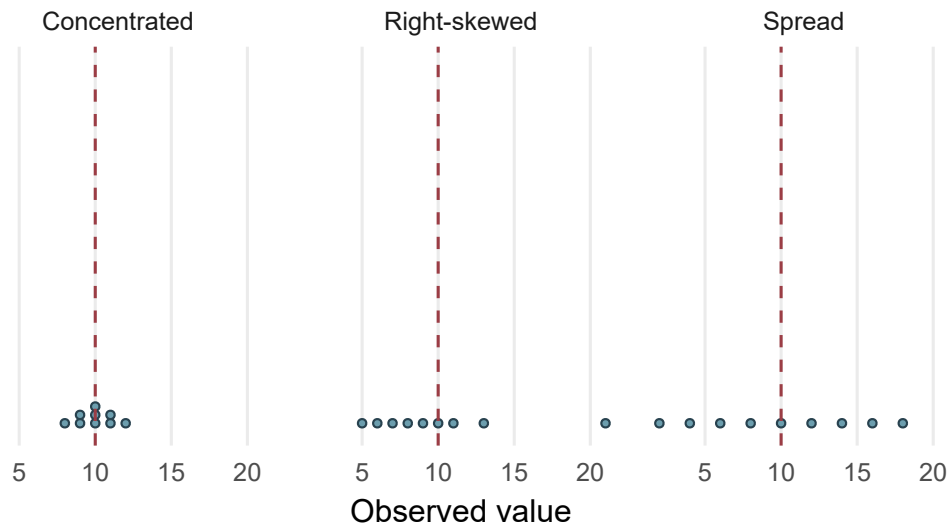


Figure 3: Three constructed distributions share a mean of 10 but differ in shape and spread.

communicate center and spread more honestly. In either case, retain a graph. Resistant summaries are not repairs for invalid measurement, selection bias, or unexplained exclusions.

3.4 A brief Normal-model and z-score bridge

The **Normal distribution** is a symmetric, bell-shaped probability model located by a mean μ and scaled by a standard deviation σ . It is not the default shape of every dataset. The **standard Normal distribution** has mean 0 and standard deviation 1.

If a modeled variable follows $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

For observed sample values, a standardized score is often written

$$z_i = \frac{x_i - \bar{x}}{s}.$$

It records how far an observed value lies from the sample mean in sample standard-deviation units. For example, a sample of commute times has mean 26 minutes and SD 6 minutes. A 38-minute commute has $z = (38 - 26)/6 = 12/6 = 2$: it is two sample standard deviations above the sample mean. Model standardization would instead use the declared model parameters

μ and σ . Subtracting a center changes location; dividing by a positive SD changes scale, not shape. Computing z scores does **not** make a skewed dataset Normal.

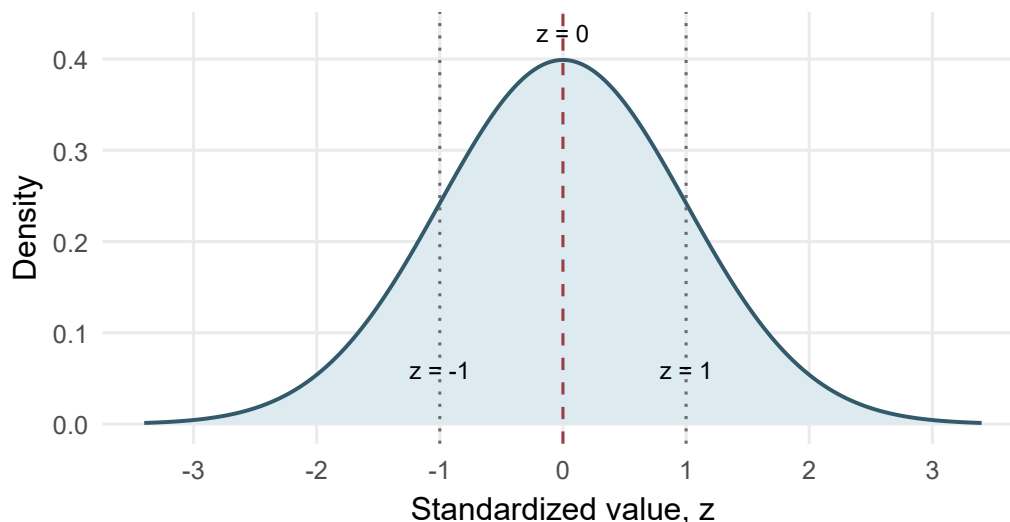


Figure 4: The standard Normal model centers distance at zero and measures it in standard-deviation units.

This is a preview of a probability distribution, not a normality test or a tail-area exercise. Section 5 distinguishes the observed data distribution from a modeled probability distribution and from a sampling distribution.

3.5 Choose a graph and summary from the question

Table 7

A graph-and-summary decision guide.

Variable forms	Useful first display	Initial graph-reading focus
One quantitative variable	Dotplot, histogram, or boxplot	What shape, center, spread, and unusual observations are visible?
Quantitative and quantitative	Scatterplot	What form, direction, strength, support, and unusual observations are visible?
Quantitative and categorical	Grouped distributions	How do distributions differ across groups?
Categorical and categorical	Counts with declared conditional percentages	Which denominator answers the comparison question?

A graph is evidence, not decoration. Numerical summaries are purposeful reductions of that evidence. A good description names what the chosen reduction reveals and what it omits.

4 See and Summarize a Relationship

A relationship asks how values of one variable are arranged with values of another. The first display follows the variable forms and the question. This section develops a quantitative–quantitative relationship because it serves the focal absence–grade question. Grouped distributions and categorical percentages are introduced later when they can receive full treatment.

4.1 Form before coefficient

Four small displays in Figure 5 show why *form* comes before a coefficient.

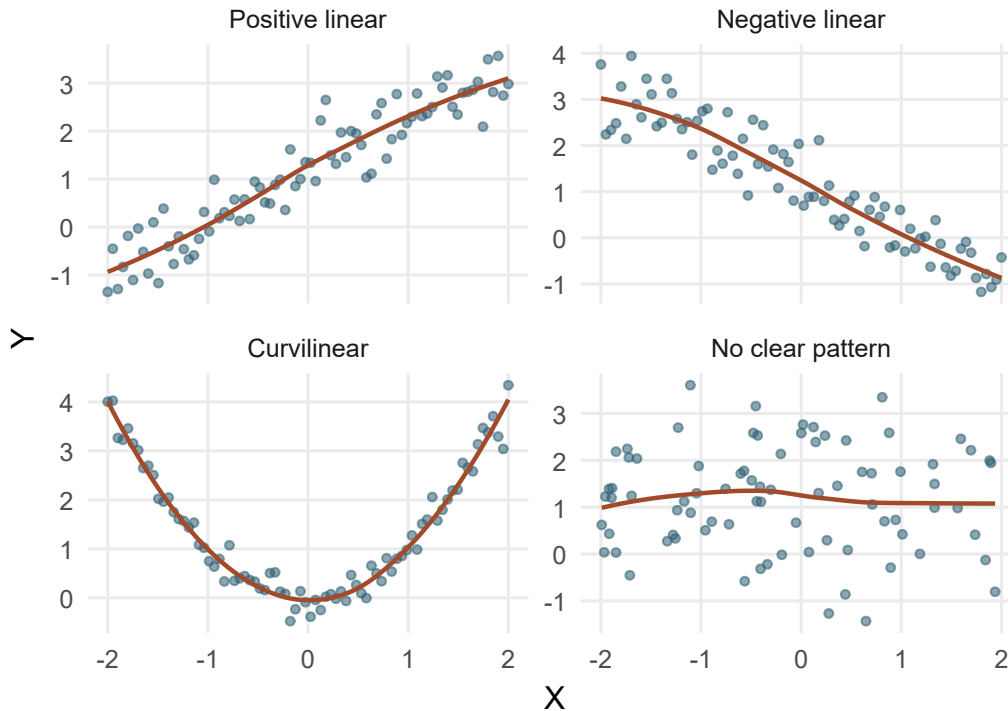


Figure 5: Linear, curvilinear, and patternless relationships require different descriptions.

Describe a scatterplot by:

1. **form:** linear, curved, clustered, or otherwise structured;
2. **direction:** increasing, decreasing, or mixed;

3. **strength**: how tightly observations follow the form;
4. **support**: where observations actually occur; and
5. **unusual observations**: points or groups that may change the summary.

Nonlinear or *curvilinear* describes form. It does not mean random, unimportant, or causal. A strong U-shaped relationship can have Pearson's correlation near zero because its negative and positive linear tendencies cancel.

4.2 Covariance and Pearson's correlation

For paired observations (x_i, y_i) , the sample covariance is

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

Positive covariance means paired deviations tend to have the same sign; negative covariance means they tend to have opposite signs. Its magnitude depends on the units of both variables. Pearson's correlation standardizes covariance:

$$r = \frac{s_{xy}}{s_x s_y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}.$$

The second form follows because the $n-1$ factors cancel. Consider four paired records with $x = (1, 3, 3, 5)$ practice hours and $y = (2, 2, 4, 4)$ score points. Both means are 3. The x deviations are $-2, 0, 0, 2$ and the y deviations are $-1, -1, 1, 1$. Their cross-products are $2, 0, 0, 2$, summing to 4. Hence

$$s_{xy} = \frac{4}{3}, \quad s_x = \sqrt{\frac{8}{3}}, \quad s_y = \sqrt{\frac{4}{3}}, \quad r = \frac{4}{\sqrt{8 \times 4}} = 0.707.$$

Covariance has hour-point units; dividing by the two SDs cancels those units. If hours become minutes, each x deviation and s_x is multiplied by 60. Covariance therefore becomes $60(4/3) = 80$ minute-points, but the same factor cancels in r , which stays .707.

Equivalently, write $z_{xi} = (x_i - \bar{x})/s_x$ and $z_{yi} = (y_i - \bar{y})/s_y$. Then

$$r = \frac{1}{n-1} \sum_{i=1}^n z_{xi} z_{yi}.$$

These are standardized cross-products, with divisor $n-1$, **not** n , because the z scores use sample SDs. Positive products support a positive linear association and negative products support a

negative one. The bound $-1 \leq r \leq 1$ follows from a property of these squared-deviation sums, not merely from averaging; Appendix C gives a short explanation. Correlation is unchanged by positive rescaling, reverses sign if one scale is reversed, and is undefined if either variable has zero SD. Independence implies zero covariance when second moments exist, but zero covariance does not imply independence: the curved example already shows why.

Table 8

Covariance and Pearson’s correlation have connected but different jobs.

Feature	Sample covariance s_{xy}	Pearson correlation r
What it summarizes	Sum of paired-deviation products divided by $n - 1$	Standardized linear direction and strength
Units	Product of the two variables’ units	None
Range	Unbounded; scale dependent	From -1 to 1
Effect of changing measurement units	Magnitude changes	Unchanged by positive linear rescaling
What it cannot reveal alone	Curvature, clusters, restricted support, or influential observations	The same omissions; near zero is not proof of no relationship

For the focal records, the sample covariance is 1.26 absence-count-by-grade-point units and Pearson’s $r = .034$. Converting absences from counts to a positively rescaled unit would change the covariance magnitude but not r . Neither number should be read without the graph.

4.3 The focal relationship

Figure 6 preserves all 395 paired observations. Jitter makes overlapping integer-valued points visible; it does not change the values used for the coefficient.

Most observations occupy low absence counts, several much larger counts extend the support, and the overall linear pattern is weak. The smooth does not reveal a strong simple curve that would overturn the near-zero linear summary, but it is a descriptive check rather than proof that no localized or subgroup pattern exists.

The concentrated zero set is consequential. Omitting the 38 unresolved zero-grade rows leaves 357 records and changes the correlation from .034 to -0.213 . The literal result is therefore not robust to this coding pattern. That sensitivity does not establish that omission is correct; it establishes that the pattern must remain visible in the answer.

A bounded statistical description is:

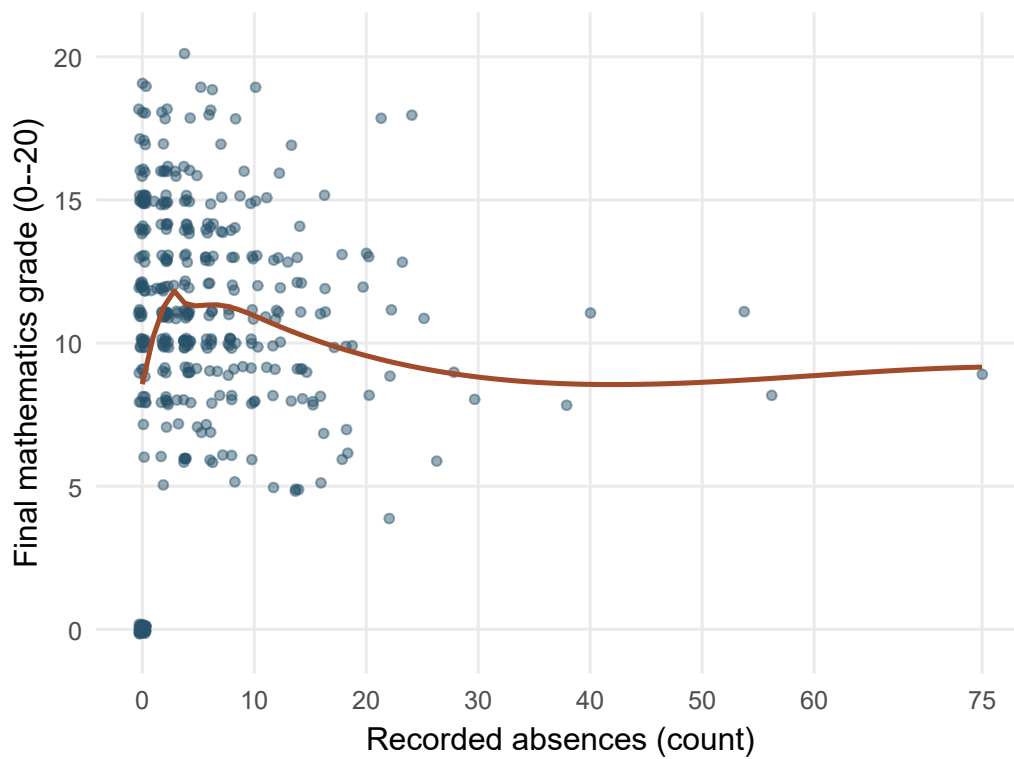


Figure 6: Recorded absences and final mathematics grade in the observational two-school record.

Across these 395 paired records, the calculated Pearson correlation between absences and final grade was $r = .034$, indicating almost no overall linear association. The scatterplot shows dense overlap at low absence counts and a few much larger counts. Omitting the 38 source-coded zero grades as a diagnostic changes the correlation to $r = -.213$; because the available documentation does not explain why every zero grade is paired with zero absences, the as-recorded coefficient does not settle a coding-insensitive direction or exclude every nonlinear or subgroup pattern.

The substantive interpretation must retain the observational design, recorded measures, and local scope:

In these two-school records as coded, absence counts and final mathematics grades show little overall linear association. A diagnostic omission of the 38 zero-grade records yields a modest negative association, so the unresolved coding context is material. Neither result establishes that attendance is unimportant, that changing attendance would leave grades unchanged, or that the same pattern holds for other students.

4.4 One estimand, complementary checks

The graph, smooth, covariance, and correlation play different supporting roles in one argument. The declared question-facing association—the Pearson correlation for the defined group of 395 records—is the estimand. The calculated correlation, $r = .034$, gives its value for the as-recorded analytic set. The graph and smooth check whether that scalar summary hides visible form or support problems. Covariance exposes the scale-dependent quantity standardized by r .

The zero-grade diagnostic changes the analytic set and therefore the finite-record quantity. It is a sensitivity check on the interpretation of the primary as-recorded estimand, not a second estimate of that identical estimand. Adopting the exclusion as the target analysis would require an explicitly revised record definition and justification.

Several displays, summaries, and checks serving the same estimand do not create additional question–answer pairs or arguments. A question can also need more than one estimand: parallel statistical strands can then contribute to its one integrated substantive answer. A separate argument is needed when the study owes a separate substantive answer to a different question, not merely because it reports another statistic. This focal question declares just one estimand.

4.5 Three necessary criteria for a causal claim

A causal interpretation must address three necessary criteria:

1. **Association.** The proposed cause and outcome differ or co-vary in the relevant comparison.
2. **Temporal order.** The proposed cause precedes the outcome over a defensible time scale.

3. **Credible handling of alternative explanations.** The design and analysis make confounding, selection, measurement, history, and other plausible explanations sufficiently implausible for the claim being made.

These are criteria, not three mechanical proofs. The focal records do not establish when every absence occurred relative to the assessed outcome and do not credibly rule out alternative explanations. Even a visible association would not be enough for a causal answer.

For example, prior academic difficulty, motivation, or illness could influence both later attendance and final performance, creating a **common-cause** explanation. Earlier low grades might instead lead to later disengagement and absence, illustrating **reverse direction**. Students represented in these two schools may also differ from students elsewhere, so **selection** can limit the scope of the pattern. These are candidate structures to investigate, not facts established by this file.

Association here means a difference or relationship in the relevant causal comparison, not necessarily a nonzero Pearson correlation or a statistically significant result. A near-zero overall correlation can hide a nonlinear relationship or different effects in subgroups; it does not rule out causation.

Random assignment is a powerful design route for the third criterion because it balances pre-assignment alternatives in expectation and establishes assignment before an outcome. It does not automatically prove faithful intervention delivery, valid measurement, harmless attrition or noncompliance, absence of interference, or transport to other settings. **Random selection** addresses population representation instead. Confusing the two changes what a study can claim.

5 From One Dataset to Repeated Samples

The focal record addresses a finite-record associational question with descriptive analysis. Many studies instead use a sample to learn about a wider population or compare outcomes under a chance-based design. Readers then need to distinguish three distributions and understand why a statistic varies across repeated samples.

5.1 Three distributions answer three questions

Table 9

Data, probability, and sampling distributions answer different questions.

Object	What varies	Question answered	Example
Data distribution	Values across observed cases	What pattern is present in these observations?	The 395 recorded grades
Probability distribution	Possible outcomes under a stated model	What could happen, and with what modeled chances?	A Normal model or a fair-die model
Sampling distribution	A statistic across comparable repeated samples or assignments	How would this statistic vary across repetitions?	Sample means from repeated random samples of size 30

A data histogram contains observed individual values. A Normal curve represents a model for possible values. A sampling distribution contains statistics, such as sample means, produced across comparable repetitions. A Normal-looking sampling distribution does not imply that the observed individual values are Normal.

A probability statement is incomplete without a generating story. The story might be random sampling, randomized assignment, repeated measurement error, or an explicitly assumed model. The arrows in Figure 7 describe such a process; they do not assert that the focal school records were randomly sampled from every population a reader can imagine.

5.2 Population, sample, parameter, statistic, and estimand

These terms describe different parts of the question and evidence. They should not be treated as interchangeable vocabulary.

Table 10

A linked family of sampling concepts.

Concept	Beginner meaning	Important distinction
Population	Defined collection of units, occasions, or outcomes about which a question is posed	Must be named; it is not automatically everyone
Sample	Units or outcomes selected or observed under a stated process	The analytic sample may be smaller than the observed record
Parameter	Fixed, usually unknown feature of a defined population or probability model	It may be the estimand when the question targets it

Concept	Beginner meaning	Important distinction
Statistic	Numerical function calculated from sample data	It becomes an estimate when used to learn about an estimand
Estimand	Precisely specified quantity the question seeks	What the study seeks, not the observed numerical answer
Estimator	Rule applied to each comparable sample to estimate an estimand	The same rule can produce different estimates
Estimate	Observed numerical result used for an estimand	One realized statistic, not its sampling distribution
Sampling distribution	Distribution of a specified statistic across comparable repetitions	Its observations are statistics, not people or scores
Sampling variability	How a statistic differs across repetitions	Repeat-to-repeat variation, not a data-entry mistake
Realized sampling error	Difference between one estimate and its estimand under the stated model	Usually unknown; not systematic bias or measurement error
Standard error	Standard deviation of a statistic's sampling distribution, or an estimate of that spread	Variability of a statistic, not spread among individuals

Suppose a defined population contains all 20,000 students enrolled in a system on a specified date. Its mean grade μ is a parameter. If the question seeks that mean, μ is also the estimand. A probability sample of 100 students produces a sample mean $\bar{x}$, a statistic used as an estimate. The **estimator** is the rule “add the selected grades and divide by 100.” Before a sample is drawn, $\bar{X}$ denotes the value that this rule will produce; after observing it, $\bar{x}$ denotes its realized value. Suppose the observed mean is 10.6 grade points. Repeating the same sample design produces many sample means, perhaps 10.6, 11.2, 10.9, and so on; their distribution is the sampling distribution. Its SD is the SE, not the SD among 100 students.

If we knew that the population mean were 11.0, the **realized sampling error** would be $\bar{x} - \mu = 10.6 - 11.0 = -0.4$ grade points. The negative sign means this estimate falls below the estimand. In practice that difference is usually unknown because μ is unknown. **Sampling variability** describes the spread across all repetitions. **Systematic bias** concerns a procedure that tends to miss the estimand in a direction. Undercoverage, nonresponse, dependence, invalid measurement, and processing mistakes are nonsampling problems; larger n does not automatically repair them.

For the model illustration that follows, suppose observations are independent draws from the same distribution, with finite mean μ and finite variance σ^2 . Its center and spread can be written

$$\mu = \mathbb{E}[X], \quad \sigma^2 = \text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

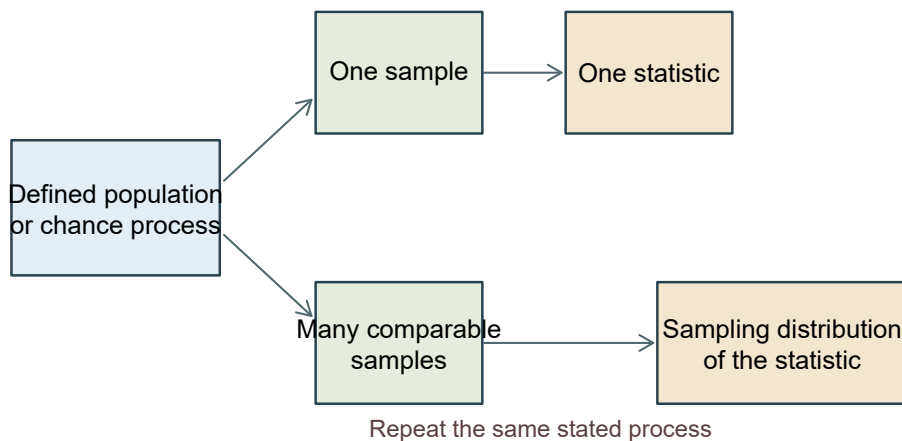


Figure 7: Repeated sampling links a defined population or chance process to a sampling distribution.

Here $\mathbb{E}$ means **expectation**, the probability-weighted average over possible outcomes, not an Evidence component such as STAT-E. Variance is the expected squared distance from the model mean. This is an explicit sampling model for teaching; it is not a newly assumed population model for the 395 school records. A simple random sample without replacement from a finite list is a different design, treated briefly in Appendix C.

We can now finish the $n - 1$ explanation. Under these independent, identically distributed, finite-variance conditions, $\mathbb{E}[s^2] = \sigma^2$: the conventional sample variance is **unbiased** for the model variance. Across repeated samples its average equals σ^2 ; a particular s^2 can be too high or too low. Dividing the squared-deviation sum by n instead has expectation $(n - 1)\sigma^2/n$ and is downward biased for that model variance. The zero-sum mnemonic and this sampling property answer different questions; Appendix C provides the short derivation (Moore et al., 2014).

5.3 Repeated sampling, standard error, and stabilization

To see repeated sampling clearly, imagine a stable right-skewed population with mean 10 and standard deviation 10. Take independent random samples of size n , calculate the mean, replace the sample, and repeat 5,000 times. Figure 8 and Table 11 summarize the resulting sample means.

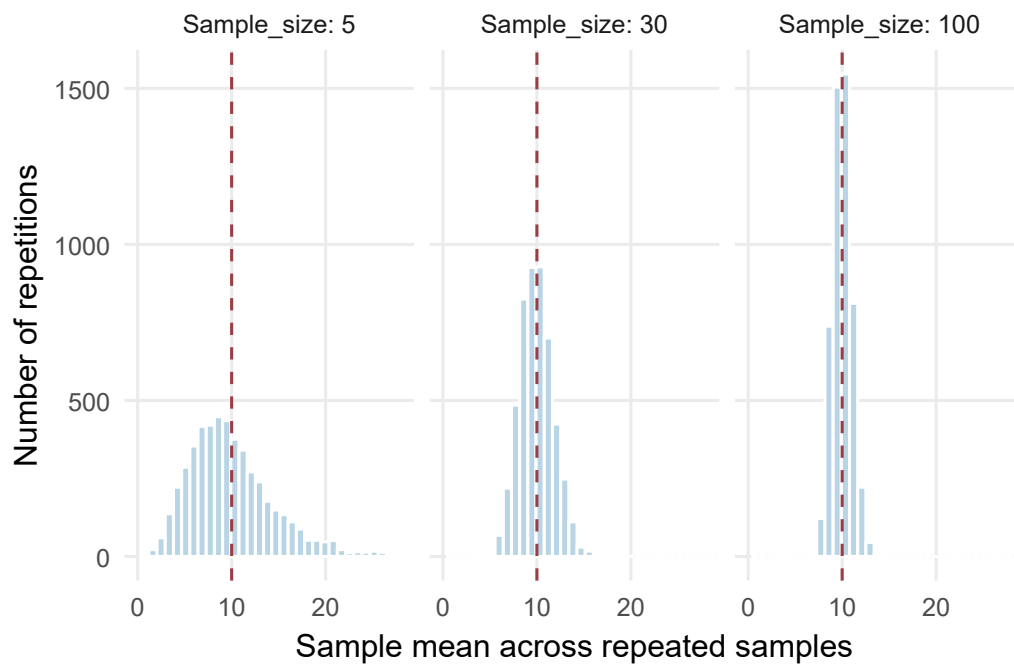


Figure 8: Sampling distributions of the mean from a right-skewed population narrow and become more symmetric as sample size grows.

Table 11*Repeated-sampling summaries under the stated right-skewed population model.*

Sample size	Model SE	Mean of sample means	Empirical SE	Skewness
5	4.47	10.01	4.54	1.00
30	1.83	9.96	1.79	0.31
100	1.00	10.00	1.01	0.22

The sample means remain centered near 10. Under the stated independent, identically distributed model, averaging n observations with expectation μ still has expectation μ . Independence makes their variances add, and dividing a sum by n divides its variance by n^2 :

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}, \quad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The **Model SE** column uses that last formula: $10/\sqrt{5} = 4.47$ and $10/\sqrt{100} = 1.00$. The **Empirical SE** column is the observed SD of the 5,000 simulated sample means. Its close, but not exact, agreement with Model SE is the intended check; a finite simulation introduces some fluctuation. Neither column is the SD of individual observations, which stays 10 in the generating model.

Quadrupling sample size halves the SE because $\sqrt{4n} = 2\sqrt{n}$. When σ is unknown, $s/\sqrt{n}$ estimates the mean's SE under these conditions. Positively related observations, such as repeated visits from one person, supply less independent information: the iid formula can then be too small. A larger row count is not a substitute for the independence assumption.

The **law of large numbers** says, under independent, identically distributed sampling with a finite mean, that the sample mean becomes increasingly likely to lie close to the population mean as n grows. In Table 11, increasing concentration around 10 is the visible lesson. The law does not force later observations to compensate for earlier ones, make individual values less variable, or turn a large convenience sample into a representative one.

The decreasing skewness in the table illustrates a different lesson: in this example, the sampling distribution of the mean becomes more nearly Normal as n grows. Under independent, identically distributed sampling with finite, nonzero variance, the **central limit theorem** explains this approximate Normal shape after centering the means and scaling by their SE. It supplies no universal sample-size cutoff. **Concentration** and **approximate shape** are not the same claim. Neither theorem repairs poor measurement, dependence, selection, or causal design.

No interval or p-value is used to convert the fixed observational school record into a population inference. These concepts prepare the repeated-sampling foundation needed when later studies develop estimators, estimates, standard errors, intervals, and tests more fully.

5.4 Complete the bounded answer and make the argument visible

Table 12 is a reading checkpoint, not an evaluation scorecard. The four rows are argument components. The linked columns show their statistical and substantive content, followed by representative documentary traces. A single source may support several components; a component may have several traces.

Table 12

Reading one associational argument, analysed descriptively, with linked statistical and substantive layers.

Component	Statistical layer	Substantive layer	Representative documentary trace and status
Evidence	395 numeric pairs for absences and final grade; visible distributions; 38 zero grades all paired with zero absences	Two-school student records; documented 0–20 final-grade scale and recorded absence counts; unexplained joint zero pattern	Adapted data file and variable documentation: located in related materials ; zero-pattern explanation: not located ; displayed pattern: generated in this chapter from that file
Justification	The declared finite-record Pearson correlation matches the requested linear association; the paired-record denominator and coding rule identify the calculation's scope	Recorded variables are interpreted as absence counts and final mathematics grade, within the documented schools and scale	Analysis rationale: stated in this chapter; source meanings: located in related materials
Inference	The plot and $r = .034$ indicate almost no overall linear association as recorded; $r = -.213$ after the diagnostic omission shows material sensitivity to the zero pattern	The numerical and graphical results are interpreted as local observational descriptions with unresolved coding context, not intervention effects	Figure and summaries: generated in this chapter; zero-pattern explanation: not located ; interpretation: stated in this chapter

Component	Statistical layer	Substantive layer	Representative documentary trace and status
Bounded Claim	Almost no overall linear association in all 395 source-coded records, with a modest negative diagnostic after omitting zero grades; no preferred deletion or exclusion of hidden forms	The as-recorded direction does not settle a coding-insensitive attendance–performance relationship; no conclusion that attendance is unimportant, causal, or population-wide	Opening overclaim: hypothetical wording introduced in this chapter; revised sensitivity-bounded answer: constructed in this chapter; no source-paper contradiction is claimed

The statistical layer establishes what was calculated and inferred from the record. The substantive layer builds on and contextualizes that support. Its Justification augments and contextualizes the statistical justification by connecting the recorded variables, school setting, and question; it does not duplicate or automatically strengthen the statistical work. The transitions must preserve magnitude, visible form, design status, and scope.

The opening school report is hypothetical, so this worked example has no source paper to which the analysis, figures, interpretation, or revised answer can be attributed. The external documentary traces used here are the adapted data file and its source documentation; the chapter generates and interprets its own analysis. In a real study, if a relevant component cannot be found, record **not located** as the documentary status. That label does not mean the work was never done. The pair-level phrase **not assessable from the accessible record** belongs to quality evaluation, not documentary tracing.

i Secondary use: selected quality-evaluation questions

Only after reading the argument do we evaluate its support. Four of Chapter 1’s six questions are especially informative here:

Quality-evaluation question	Judgment and reason
Evidence integrity	Insufficient information to judge. The underlying measurement cannot be settled: numeric entries, documented ranges, and the denominator are traceable, but the sources do not explain why all 38 final-grade zeros pair with zero absences. The as-recorded analysis and sensitivity preserve rather than resolve that limitation, and neither establishes wider representation.
Inferential adequacy	Adequate for this purpose. For the as-recorded estimand after reporting sensitivity, the graph, smooth, covariance, and primary correlation address visible form and the declared all-record quantity; the diagnostic omission shows that a coding-insensitive interpretation remains unsupported.
Transition fidelity	Needs revision. The opening move from near-zero recorded correlation to what attendance “matters” for drops the observational and local limits.
Claim calibration	Needs revision. “Attendance does not matter” is stronger and broader than a near-zero finite-record linear association supports.
These judgments form a profile, not a summed score. The pair-level evaluation conclusion is supported only after narrowing or revision: the same associational question can be answered defensibly by replacing the opening overclaim with the bounded local answer above, provided that the unresolved zero-pattern sensitivity remains attached.	

i Auxiliary use: selected construction stages

For a researcher conducting a similar study, this chapter illuminates parts of Stages 3–6:

- **Stage 3:** Align design, measurement, estimand, evidence-production or acquisition plan, and inferential plan.
- **Stage 4:** Produce or acquire, and document, empirical and contextual evidence.
- **Stage 5:** Estimate, check, and analyse.
- **Stage 6:** Contextualize, interpret, and complete the bounded answer.

Here Stage 5 includes the graph-first description.

Adding jitter or a complementary display after checking the data is a within-Phase-2 repair: the Phase 1 anchor has not changed. Retroactively changing the question to a causal effect changes the anchored question, mode, estimand, and design obligation. The existing cycle must receive an explicit outcome, and an additional or revised cycle is required. Defensible elements may be reused only with their provenance and changed status visible.

6 Summary and Reading Bridge

A careful reading of an associational question analysed descriptively follows the support from question to bounded answer:

1. State the question, estimand, and intended answer mode.
2. Identify what each row, variable, value, scale, and denominator represents.
3. See the observations before reducing them.
4. Choose graphs and summaries that fit the question, variable forms, and visible shape.
5. Distinguish linear association from curvilinear form and from causation.
6. Distinguish a statistic in one sample from its sampling distribution.
7. Carry measurement, design, uncertainty, and scope limits into the bounded answer.

For the focal record, the answer is modest: absences and final grades show almost no overall linear association among all 395 source-coded records, while a diagnostic omission of the 38 zero grades gives a modest negative association. Because the documentation does not explain that joint zero pattern, neither result licenses the stronger claims that attendance is unimportant, that changing attendance would not affect grades, or that the same result holds elsewhere.

Reading the argument is the primary use. Evaluating its quality is a dependent secondary use, illustrated by four selected questions. The construction workflow is an auxiliary use for researchers and was invoked only where the chapter's descriptive work illuminates Stages 3–6.

6.1 Essential glossary

Case. The entity or occasion represented by one row.

Distribution. The arrangement of values across observed cases, possible model outcomes, or repeated statistics, depending on the object named.

Estimand. The precisely specified quantity the question seeks.

Estimate. The observed numerical result used to learn about an estimand.

IQR. The distance $Q_3 - Q_1$, covering the middle half of ordered observations.

Pearson’s correlation. A unit-free summary of linear direction and strength between two quantitative variables.

Sample variance and standard deviation. Squared and original-unit measures, respectively, of spread around the sample mean.

Sampling distribution. The distribution of a specified statistic over comparable repeated samples or assignments.

Standard error. The standard deviation of a statistic’s sampling distribution, or an estimate of that spread.

6.2 Further reading

- The ASA GAISE College Report emphasizes statistical problem solving through questions, data, analysis, and interpretation (American Statistical Association, 2016).
- OpenIntro’s free materials connect data graphics, numerical summaries, and foundations for inference (Çetinkaya-Rundel & Hardin, 2024).
- NIST’s exploratory-data guidance emphasizes graphical examination before model reduction.
- The UCI student-performance source documents the variables and provenance of the worked record (Cortez, 2008).

Exercises

The exercises are modular and code independent. Use only the evidence provided unless a prompt explicitly invites outside material.

Exercise 1: Identify the evidence structure (Foundational)

A clinic report contains one row per visit. Some patients appear in several rows. The file records patient ID, clinic, visit date, waiting time in minutes, and whether the patient returned within 30 days. Of 1,240 visits, waiting time is missing for 36, return status is missing for 81, and 10 visits are missing both.

Identify:

1. the case represented by one row;
2. the observational unit or occasion;
3. one identifier, one quantitative variable, and one categorical variable;
4. why 1,240 visits should not be described as 1,240 independent patients;
5. the complete-case denominator for an analysis using waiting time and return status; and
6. one question that must be answered before generalizing beyond these visits.

Exercise 2: See and summarize one distribution (Foundational)

Eight recorded completion times, in minutes, are:

2, 4, 4, 4, 5, 5, 7, 9

1. Sketch a dotplot or ordered display.
2. Calculate the mean, median, sample variance, sample standard deviation, and IQR. For quartiles, use the medians of the lower and upper halves.
3. State the units of the variance and standard deviation.
4. If the question asks for a resistant description of typical time and spread, choose a pair of summaries and explain why.
5. Name one feature that the chosen pair does not show.

Exercise 3: Read a Normal model and standardized value (Foundational)

A reference model for a score is Normal with mean 50 and standard deviation 10. A learner receives 65.

1. Calculate and interpret the learner's z score.
2. Distinguish this model-based standardization from using a sample mean and sample standard deviation.
3. Explain why transforming every observed score to z scores would not make a skewed observed distribution Normal.
4. State one question that cannot be answered without additional probability information or a Normal table.

Exercise 4: Compare a graph, covariance, and correlation (Foundational)

A scatterplot of weekly practice hours and assessment score follows a clear U-shaped curve. The sample covariance is 0.80 hour-points and Pearson's correlation is .03.

1. Why can the graph show a strong relationship while r is near zero?
2. What does the positive sign of the covariance summarize?
3. Why is the covariance magnitude not directly comparable across measurement units?
4. If hours are converted to minutes, what happens to the covariance and to Pearson's correlation?
5. Write one bounded descriptive sentence that respects the graph.

Exercise 5: Apply the three causal criteria (Integrative)

An observational study finds that students who voluntarily attend tutoring have higher end-of-term scores than students who do not. Attendance is recorded during the term; the outcome is measured at the end.

1. Which of the three necessary causal criteria are partly visible from this information?
2. Give two plausible alternative explanations.
3. Explain how random assignment to a tutoring offer could strengthen the design.
4. Name two problems random assignment would not automatically solve.
5. Write an associational answer supported by the stated evidence.

Exercise 6: Interpret population, sample, and repeated sampling (Integrative)

A school system defines its population as all 20,000 enrolled students on 1 September. It draws a simple random sample of 100 students without replacement and uses their mean attendance rate to estimate the population mean. Across many such samples, the mean of the sample means is 91.0%, their standard deviation is 1.2 percentage points, and the realized sample mean is 89.7%.

1. Identify the population, sample, parameter, statistic, estimand, estimator, estimate, sampling distribution, sampling variability, realized sampling error, and standard error.
2. If the true population mean is 91.0%, calculate the realized sampling error.
3. If the sample size increased from 100 to 400, use the square-root rule to approximate the new SE. Explain why this is an approximation for sampling without replacement, rather than an exact independent-sampling result.
4. State why a larger sample would not repair an invalid attendance measure or an incomplete sampling frame.
5. Relate the increasing concentration of sample means to the law of large numbers without using a compensation story.

Exercise 7: Read and then evaluate a bounded argument (Integrative)

A report asks: “Among 260 volunteers in an online training course, how strong is the linear relationship between logged study minutes and final quiz score?” It documents one row per volunteer, 18 missing quiz scores, a Pearson correlation of .41 among 242 complete pairs, and a scatterplot showing a moderate positive linear pattern with two high-leverage observations. The report concludes: “More study time raises scores for online learners everywhere.” The course platform’s data dictionary is available, but the recruitment notice and analysis script are not located.

1. State the question–answer pair and its intended estimand for all 260 volunteers, preserving what the report actually asks and claims.
2. Fill the Evidence, Justification, Inference, and bounded Claim rows for the statistical and substantive layers.
3. Identify representative documentary traces and give each a documentary status. Record any contradiction separately.
4. State the report’s answer before judging it.
5. Apply Evidence integrity, Inferential adequacy, Transition fidelity, and Claim calibration using Chapter 1’s judgment labels. At this evaluation stage, explain whether the complete-pair correlation identifies the 260-volunteer estimand without additional assumptions.
6. Give one pair-level evaluation conclusion for the original 260-volunteer pair and state what, if anything, the accessible record supports. If you formulate an answer for the 242 complete pairs, state whether it belongs to the original pair or to an additional or revised pair.
7. Is checking the two high-leverage observations a within-Phase-2 repair, or does it require an additional or revised cycle? Explain.
8. Would changing the question to the causal effect of assigning extra study time preserve the Phase 1 anchor? Explain.

Appendix A: Conditional Probability and Independence

Conditional probability changes the reference set. For events A and B , $A \cap B$ means “both A and B occur,” and $P(A \mid B)$ means “the probability of A given that B occurred.”

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

The denominator is the number or probability of cases satisfying B , not the whole record. Suppose a study records the following support and pass counts:

Table 13

Hypothetical two-way display of support and pass status for denominator practice.

Recorded school support	Passed	Did not pass	Total
Yes	14	37	51
No	218	126	344
Total	232	163	395

The joint empirical proportion for support and passing is $14/395 = .0354$ (3.54%). The conditional proportion passing among support recipients is $14/51 = .2745$ (27.45%). The marginal proportion passing is $232/395 = .5873$ (58.73%). Each answers a different question. These are empirical probabilities for a uniformly selected record from this table, not automatically population probabilities.

Events are independent in a probability model when

$$P(A \cap B) = P(A)P(B),$$

or, when the conditioning probability is positive, $P(A | B) = P(A)$. An observed difference between proportions is descriptive. Whether it reflects chance, bias, confounding, or a stable population difference requires further support. Empirical independence in one finite table is also different from assuming independence in a probability model.

Appendix B: Random Variables and Probability-Model Detail

A **random variable** assigns a number to each outcome of a stated chance process. A probability model should name its repeatable story, possible outcomes, parameters, and conditions.

For example, let X be the number of successful submissions in 10 hypothetical independent attempts, each with constant success probability $p = .70$. Then

$$X \sim \text{Binomial}(10, .70), \quad P(X = x) = \binom{10}{x} (.70)^x (.30)^{10-x}.$$

The expression $\binom{10}{x}$ counts the different ways to place x successes among 10 attempts. For exactly nine successes, any one attempt could be the failure, giving 10 possible arrangements: $P(X = 9) = 10(.70)^9(.30) = .1211$, about 12.1%. For this model, $\mathbb{E}[X] = 10(.70) = 7$ and $\text{Var}(X) = 10(.70)(.30) = 2.1$, so its SD is $\sqrt{2.1} = 1.45$ successes. The expectation is a model average, not a promise that a particular set of 10 attempts contains seven successes.

The model is plausible only if the number of attempts is fixed, each outcome is meaningfully binary, trials are independent for the purpose at hand, and the success probability remains constant. Real repeated attempts may violate these conditions through learning, fatigue, feedback, or shared conditions. Simulation can expose the implications of assumptions; it cannot validate the assumptions from the data alone.

A Normal model likewise has a story and parameters. Its symmetric density may be useful for a measurement model or sampling approximation, but the density does not prove that observed values are Normal. Tail probabilities answer model-based questions only after the model and its conditions are defended.

Appendix C: Stabilization, Sampling Shape, and Formal Results

The main text used one repeated-sampling table to distinguish concentration from approximate sampling shape. A running mean gives another view of stabilization.

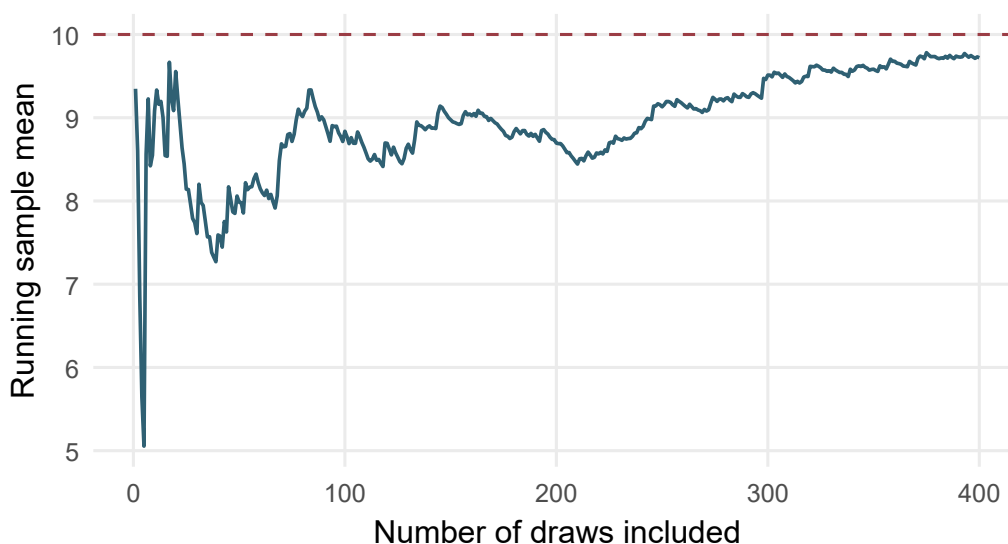


Figure 9: One running mean from independent right-skewed draws stabilizes near the population mean of 10.

Under independent, identically distributed sampling with a finite absolute mean ($\mathbb{E}[|X|] < \infty$), the law of large numbers formalizes the concentration of $\bar{X}_n$ near μ as n increases: $\bar{X}_n \xrightarrow{p} \mu$. The arrow means **convergence in probability**: for any fixed positive tolerance, the chance of falling outside it tends to zero as n grows. It does not say that short-run imbalances must reverse.

Under common conditions including finite nonzero variance σ^2 , the central limit theorem describes the standardized sampling shape:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

The arrow means **convergence in distribution**: the centered mean, expressed in its own SE units, approaches a standard Normal distribution as n grows. This is the sampling analogue of a z score, not a transformation of the raw data into a Normal population. The exact conditions and the quality of an approximation matter. Dependence, very heavy tails, complex sampling, or a statistic other than the mean can require different results.

Here **standardized** means mathematical centering and scaling used to compare sampling shapes. It is unrelated to the substantive standardization choices sometimes used in regression. Neither theorem repairs selection, dependence, measurement, processing, or causal-design weaknesses.

Why the sample-variance correction works

For independent observations $X_1, \dots, X_n$ with common mean μ and variance σ^2 , the squared-deviation decomposition is

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Taking expectations gives $n\sigma^2 - n(\sigma^2/n) = (n-1)\sigma^2$. The first term sums n expected squared distances from the population mean; the second uses the variance of the unbiased sample mean. Dividing by $n-1$ therefore gives $\mathbb{E}[s^2] = \sigma^2$. This is the expectation calculation that the zero-sum mnemonic alone does not provide.

The corresponding observed-data identity, for any proposed center a , is $\sum_i (x_i - a)^2 = \sum_i (x_i - \bar{x})^2 + n(\bar{x} - a)^2$. Its final term is nonnegative and is zero at $a = \bar{x}$. Thus the mean minimizes the sum of squared deviations: the **least-squares center**. This property prepares the logic of regression without requiring a population model.

Why correlation is bounded

When both sample SDs are positive, $\sum z_{xi}^2 = \sum z_{yi}^2 = n-1$. Because a sum of squared numbers cannot be negative, $\sum_i (z_{xi} - z_{yi})^2 = 2(n-1)(1-r) \geq 0$, giving $r \leq 1$. Likewise, $\sum_i (z_{xi} + z_{yi})^2 = 2(n-1)(1+r) \geq 0$, giving $r \geq -1$. Equality requires perfectly aligned or reversed standardized observations.

When the independent-sampling SE changes

For observations with finite variances, the more general expression is

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \left[\sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \right].$$

The second sum runs over each distinct pair of observations once. Independence makes all of those covariance terms zero. Positive total covariance increases the variance relative to the independent case; repeated records from the same person can therefore yield less precision than their row count suggests.

Sampling without replacement from a fixed list is another case. For a simple random sample of n units from N units, let S_N^2 be the list's squared deviations from its mean divided by $N - 1$. Then

$$SE(\bar{X}) = \sqrt{1 - \frac{n}{N}} \frac{S_N}{\sqrt{n}}.$$

The square-root factor is the **finite-population correction**: uncertainty vanishes when every unit is observed. It is unrelated to the $n - 1$ denominator in the sample variance. With 100 of 20,000 students, the factor is $\sqrt{1 - .005} = .9975$; with 400, it is $\sqrt{1 - .02} = .9899$. Ignoring these small corrections makes the simple square-root rule a close approximation here, but not an exact statement for the without-replacement design (Moore et al., 2014).

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